

A 30-year-old woman is seen at the preoperative evaluation clinic 2 weeks prior to her scheduled laparoscopic resection of right adrenal mass. Her initial presentation was 6-month history of hypertension and paroxysmal episodes of palpitations, dizziness, blurring of vision and headache.

A. Describe your pre-operative management of this patient's medical condition (40%).

B. List the main concerns during the intra-operative period and outline your management to address these concerns (60%).

A.

Based on her presenting history, I am concerned that this right adrenal mass represents pheochromocytoma. My pre-operative management of this patient's medical condition would be targeted at:

Assessment of disease control

- Diagnosis of pheochromocytoma should be confirmed
 - o Revisit presenting history.
 - o Biochemically on 24H urinary metanephrine study, followed by
 - o Additional radiological imaging to ensure no bilateral or extra-adrenal involvement
 - o Exclude other causes for young hypertension/ other symptoms.
- Check for associated syndromes: MEN 2A/ 2B, neurofibromatosis, von-hippel lindau syndrome
- Assess for and optimise complications associated with pheochromocytoma.
 - o Hypertensive heart disease (ECG, TTE, CXR)
 - o Stroke (neurological exam, neuro imaging as appropriate)
 - o Nephropathy (renal panel)
 - o Hyperglycaemia (BSL trend)

Medical optimisation

- Discuss with surgeons on op date (to allow adequate time ~2 weeks for medical treatment)
- Alpha adrenergic blockade (Phenoxybenzamine), followed by beta blockers (propranolol) as required.
- High sodium and fluid intake to reverse catecholamine-induced blood volume contraction.
- Admission one day prior to surgery for monitoring of BP, HR, ECG, BSL + pre-op tests including ensuring valid GXM.
- IV maintenance fluids when NBM.
- Ensure patient fulfils Roizen criteria prior to surgery:
 - o No BP > 160/90 within 24H pre-op
 - o Presence of orthostatic hypotension, with no BP < 80/45.
 - o No ST-T wave changes on ECG
- Discuss with endocrinologist on omission of phenoxybenzamine (long acting a-blocker) on morning of op to minimise risk of intra-op hypotension post-resection.

I would also perform anaesthetic assessment and counselling of anaesthesia plan including:

- Airway management
- Invasive lines
- Analgesia options
- Risks of common side effects and potential complications
- Post-op disposition (HD)

B.

During the intra-operative period, my main concerns are:

1. Peri-induction haemodynamic instability

I will address this by

- Setting invasive lines IA, CVC, Flotrac (CO output monitor with SVV, SVRI, CO/CI) pre-induction
- Avoiding histamine releasing agents
 - o Use Rocuronium and Sugammadex instead of Suxamethonium/ Atracurium/ neostigmine
 - o Use Oxycodone instead of Morphine
- Standby resuscitation drugs
 - o Noradrenaline infusion primed and connected to CVC
 - o Adrenaline 1:1000 and 1:10000 syringes drawn

2. Catecholamine surges in response to stimulus

- Anxiety
- Laryngoscopy/ intubation
- Surgical incision
- Pneumoperitoneum

I will address this by

- Premeds (Midazolam) if patient is anxious
- Video laryngoscope to reduce stimulation from forceful lifting of epiglottis during laryngoscopy
- Avoiding Desflurane and Ketamine (associated with sympathetic stimulation)
- Ensuring adequate depth of anaesthesia (titrated according to BIS)
 - o Using TCI Propofol or Sevoflurane.
- Ensuring adequate analgesia
 - o Remifentanyl infusion titrated to effect
 - o ESP/ TAP block
 - o Multimodal analgesia incl Paracetamol, Parecoxib, Oxycodone
- Running MgSO₄ infusion to inhibit catecholamine release, and for vasodilation, analgesia.
- Standby drugs
 - o Sodium nitroprusside (SNP) primed and connected to CVC
 - o Esmolol, Phentolamine syringes drawn for standby.
- Communicating with surgeons for slow controlled initiation of pneumoperitoneum

3. Hypertensive crisis/ arrhythmias during tumour manipulation/ tissue dissection

I will address this by

- Maintaining close communication with surgeons
- Titrating SNP infusion, esmolol and phentolamine boluses during this phase
- Standby amiodarone/ lignocaine in the event of arrhythmias

4. Bleeding from vessel injury

- While uncommon, this can be challenging to manage and may require conversion to open.

I will address this by

- Maintaining close communication with surgeons
- Close monitoring of suction bottles and surgical field.
- Ensuring valid GXM, standby blood products.
- Having 2x large bore IV access. Use of hotline.
- Avoid lethal triad – warm patient, adjust minute ventilation to avoid acidosis, correct coagulopathy if present.

5. Hypotension post-resection/ clamping of adrenal vessels

I will address this by

- Omit long acting alpha-blocker (Phenoxybenzamine) on day of surgery
- Fluid loading to ensure adequate preload, followed by titration to SVV < 14 after induction/ intubation
- Starting and titrating Noradrenaline infusion upon clamping of vessels.

6. Hypocortisolism post-resection

- More common in bilateral resection

I will address this by

- Administration of Hydrocortisone post-resection (after discussion with surgeon/ endocrinologist)

7. Hypoglycemia post-resection/ clamping of adrenal vessels

- Due to drop in circulating catecholamines

I will address this by

- Intra-op monitoring of POCT glucose (with ABG)
- Post-op check of capillary blood glucose in PACU and in ward.